

Objects, ThickPoint

CSE 114 INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING



ThickPoint

A mathematical point in the x,y plane has no height or width, but if you want to draw or print a point on paper or on a monitor it has to have nonzero thickness.

Write a class ThickPoint. Start by copying the file Point.java and renaming the file and the class ThickPoint, then modify the code.

Soon, we will learn how to create one class based on another class without copying it and modifying it. For now, copying and modifying is fine.

Constructor

The ThickPoint class must have this constructor:

```
/*  
    Construct ThickPoint with given x, y coordinates  
    and radius r.  
    If the point is drawn, it will be drawn as a filled circle  
    of radius r centered at the point (x,y).  
*/  
public ThickPoint(double x, double y, double r)
```

overlap and touch

You must implement these methods:

```
/*  
    Returns true if this ThickPoint and the ThickPoint p  
    will overlap when drawn.  
*/  
public boolean overlap(ThickPoint p)
```

For example the point (0, 0) with radius 1 will overlap the point (1.5, 0) with radius 1.

```
/*  
    Returns true if this ThickPoint and the ThickPoint p  
    will touch edges when drawn.  
*/  
public boolean touch(ThickPoint p)
```

For example the point (0, 0) with radius 1 will touch the point (2, 0) with radius 1.

equals and toString

Can you also implement *equals* and *toString* for ThickPoint?

If you have extra time, start to design a class that represents cars.

What instance variables should it have? (make, model, year, size, weight, ...?)

What methods should it have?

This is a free-form thought-design exercise to get you thinking about designing objects.